

MODULE III - SYNTAX PARSING

CONTENT

- Context-Free Grammars for English
- Context-Free Rules and Trees
- Noun Phrase
- Verb Phrase and Subcategorization
- Grammar Equivalence & Normal Form
- Finite State & Context-Free Grammars

SYNTAX

- By grammar, or syntax, we have in mind the kind of implicit knowledge of your native language that you had mastered by the time you were 3 years old without explicit instruction.
- Not the kind of stuff you were later taught in “grammar” school.
- Why should you care?
- Grammars (and parsing) are key components in many applications

CONTD...

- Grammar checkers
- Dialogue management
- Question answering
- Information extraction
- Machine translation

CONSTITUENCY

- A sequence of words that acts as a single unit
 - Noun phrases
 - Verb phrases
- These units form coherent classes that behave in similar ways
- For example, we can say that noun phrases can come before verbs

CONSTITUENCY

- For example, following are all noun phrases in English...

Harry the Horse
the Broadway coppers
they

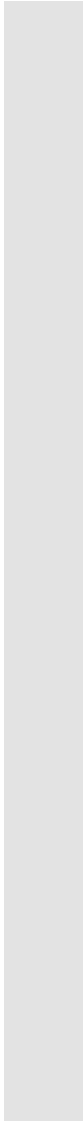
a high-class spot such as Mindy's
the reason he comes into the Hot Box
three parties from Brooklyn

Context-free grammars (CFGs) also known as

- Phrase structure grammars
- Backus-Naur form

Consist of

- Rules
- Terminals
- Non-terminals



CONTEXT-FREE RULES AND TREES

CONTEXT- FREE GRAMMARS

- Terminals
 - words
- Non-Terminals
 - The constituents in a language
 - Such as noun phrases, verb phrases and sentences.
- Rules
 - Rules are equations that consist of a single non-terminal on the left and any number of terminals and nonterminal on the right.

SOME NP RULES

- some rules for our noun phrases

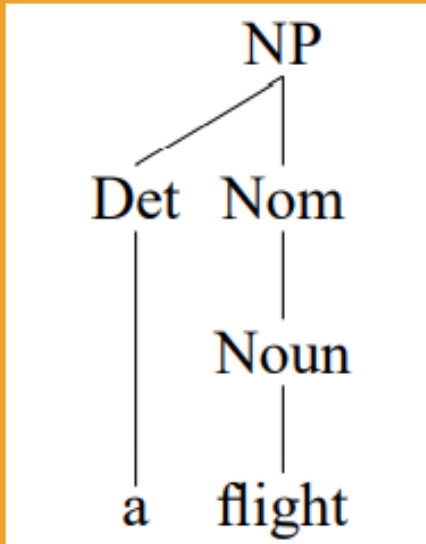
$NP \rightarrow Det\ Nominal$

$NP \rightarrow ProperNoun$

$Nominal \rightarrow Noun \mid Nominal\ Noun$

- Together, these describe two kinds of NPs.
 - One that consists of a determiner followed by a nominal
 - And another that says that proper names are NPs.
 - The third rule illustrates two things:
 - An explicit disjunction
 - A recursive definition

A parse tree
for 'a flight'.



A sentence can consist of a noun phrase and a verb phrase:

S - NP VP

I prefer a morning flight

one kind of verb phrase consists of a verb followed by a noun phrase:

VP - Verb NP

prefer a morning flight

the verb phrase may have a noun phrase and a prepositional phrase:

VP - Verb NP PP

leave Boston in the morning

CONTD...

The verb may be followed just by a preposition phrase:

VP - Verb PP

leaving on Thursday

A prepositional phrase generally has a preposition followed by a noun phrase.

For example, a very common type of prepositional phrase in the ATIS corpus is used to indicate location or direction:

PP - Preposition NP

from Los Angeles

CONTD...

- The NP inside a PP need not be a location;
- PPs are often used with times and dates, and with other nouns as well; they can be arbitrarily complex.
- Ten examples from the ATIS corpus:

to Seattle	on these flights
in Minneapolis	about the ground transportation in Chicago
on Wednesday	of the round trip flight on United Airlines
in the evening	of the AP fifty seven flight
on the ninth of July	with a stopover in Nashville

LEXICON FOR GRAMMAR RULE LO

Noun → *flights* | *breeze* | *trip* | *morning* | ...

Verb → *is* | *prefer* | *like* | *need* | *want* | *fly*

Adjective → *cheapest* | *non – stop* | *first* | *latest*
| *other* | *direct* | ...

Pronoun → *me* | *I* | *you* | *it* | ...

Proper-Noun → *Alaska* | *Baltimore* | *Los Angeles*
| *Chicago* | *United* | *American* | ...

Determiner → *the* | *a* | *an* | *this* | *these* | *that* | ...

Preposition → *from* | *to* | *on* | *near* | ...

Conjunction → *and* | *or* | *but* | ...

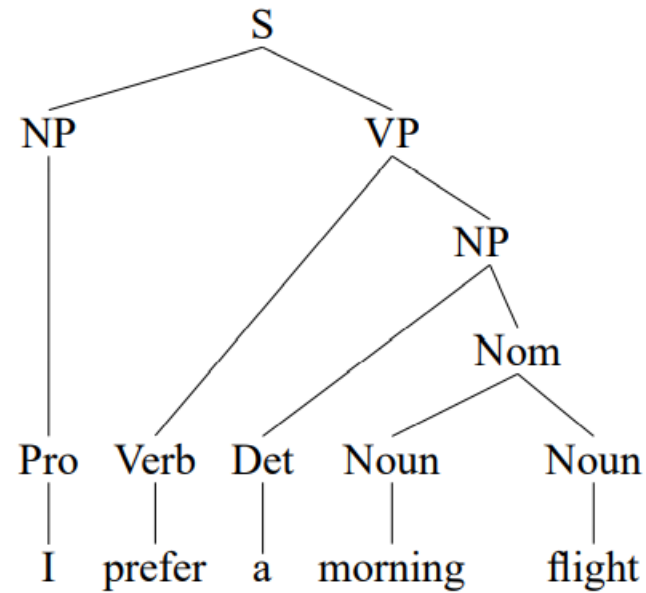
LO GRAMMAR

$S \rightarrow NP VP$	I + want a morning flight
$NP \rightarrow$	I
$Pronoun$	Los Angeles
$Proper-Noun$	a + flight
$Det Nominal$	morning + flight
Nominal $\rightarrow Noun Nominal$	flights
$Noun$	
$VP \rightarrow Verb$	do
$Verb NP$	want + a flight
$Verb NP PP$	leave + Boston + in the morning
$Verb PP$	leaving + on Thursday
$PP \rightarrow Preposition NP$	from + Los Angeles

Figure 9.3 The grammar for $\mathcal{L}_0$, with example phrases for each rule.

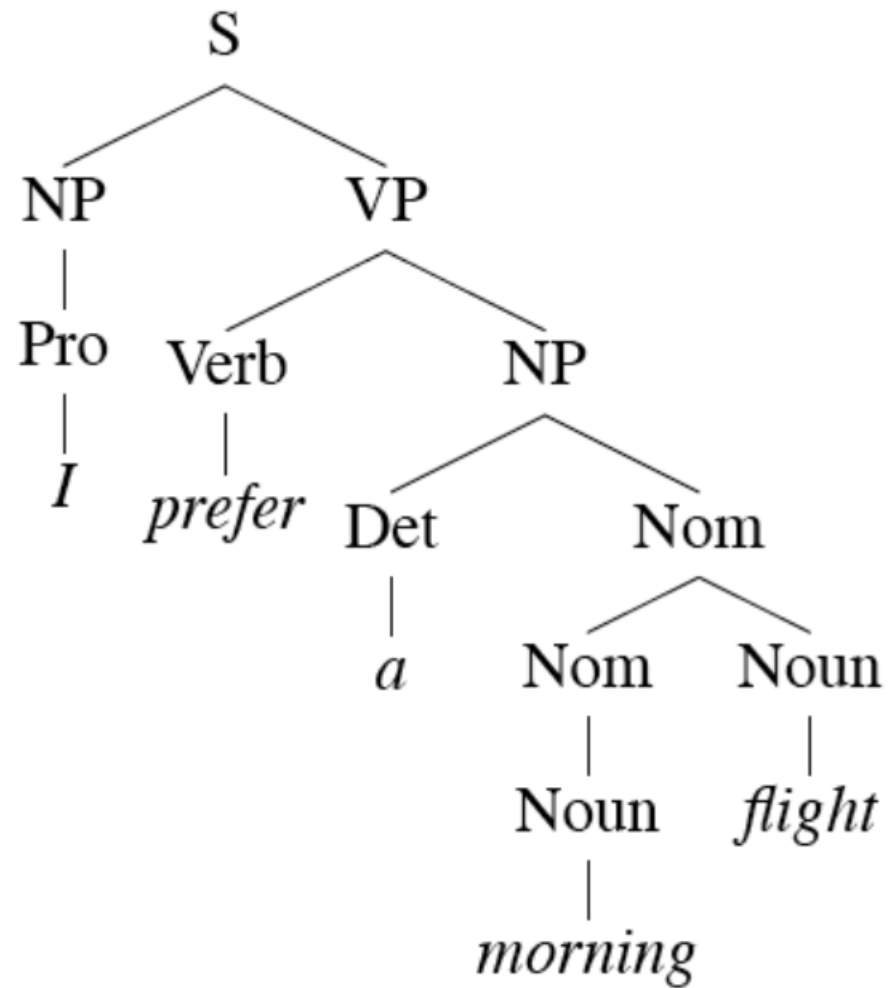
PARSE TREE FOR

‘I prefer a
morning flight’
according to
grammar L0



DERIVATION

A “derivation” is a sequence of rules applied to a string that accounts for that string.



More formally, a CFG consists of

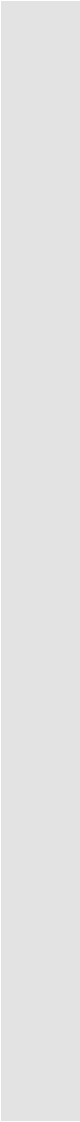
N a set of **non-terminal symbols** (or **variables**)

Σ a set of **terminal symbols** (disjoint from N)

R a set of **rules** or productions, each of the form $A \rightarrow \beta$,
where A is a non-terminal,

β is a string of symbols from the infinite set of strings $(\Sigma \cup N)^*$

S a designated **start symbol**

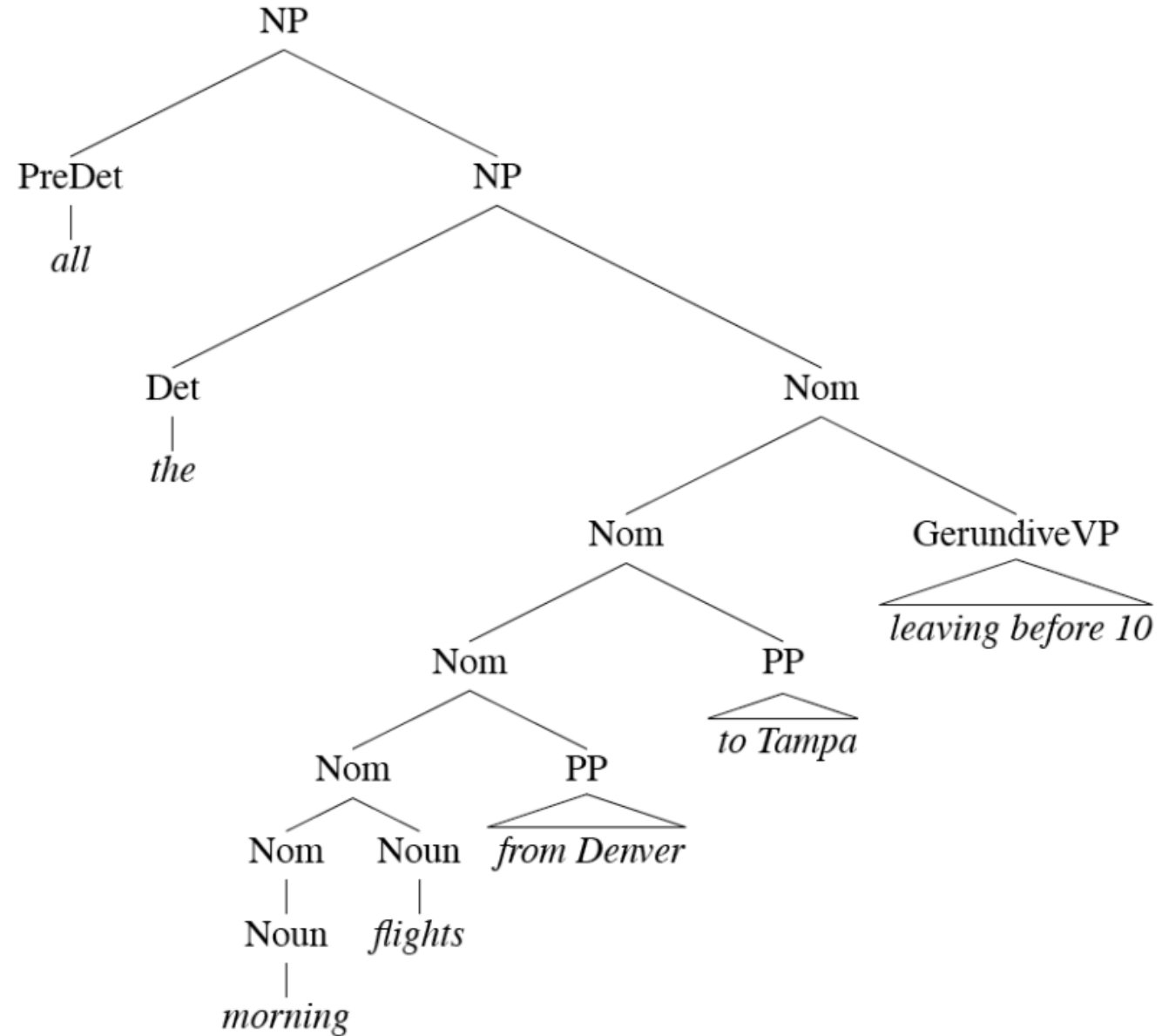


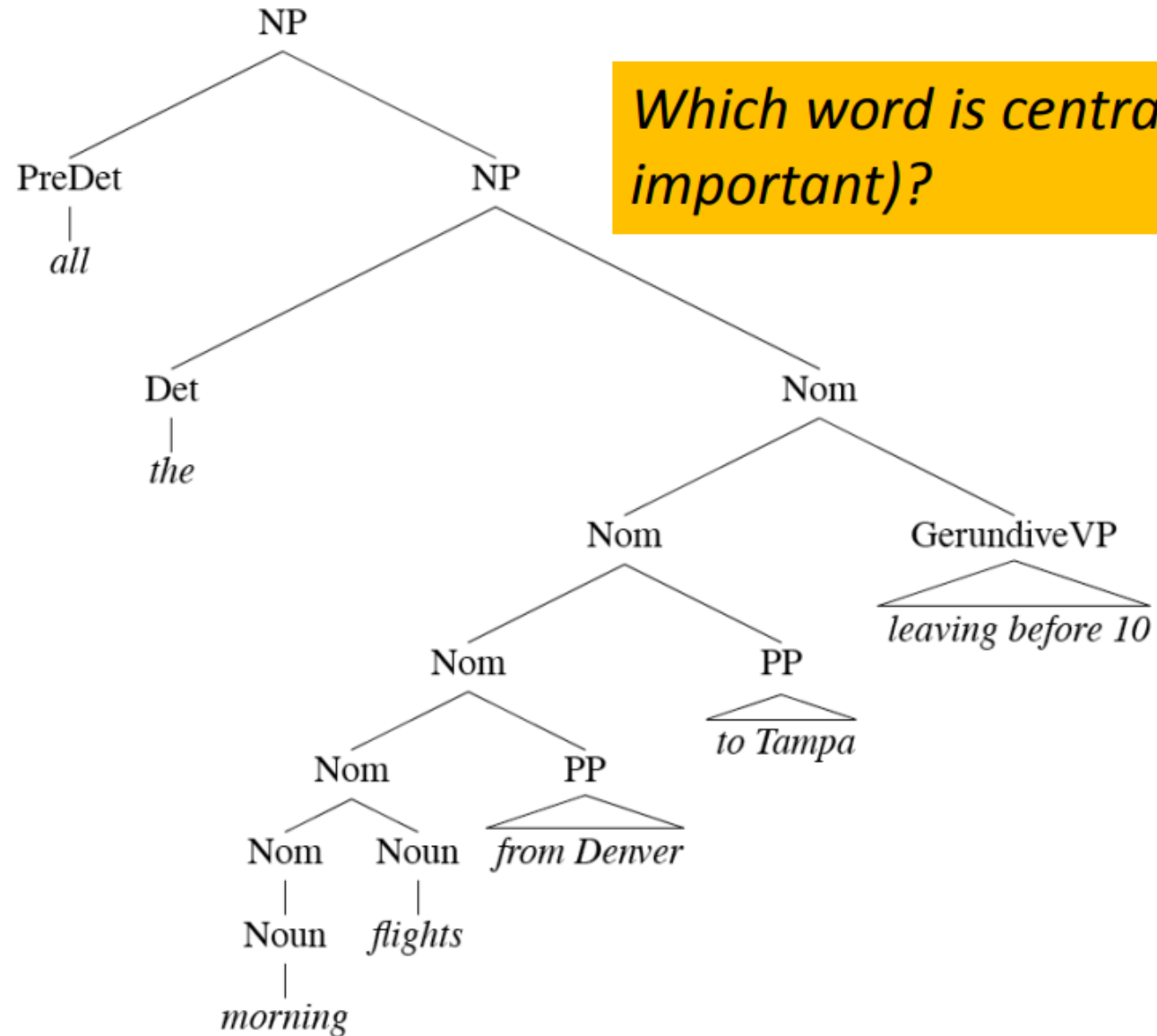
NOUN PHRASE

PARSING

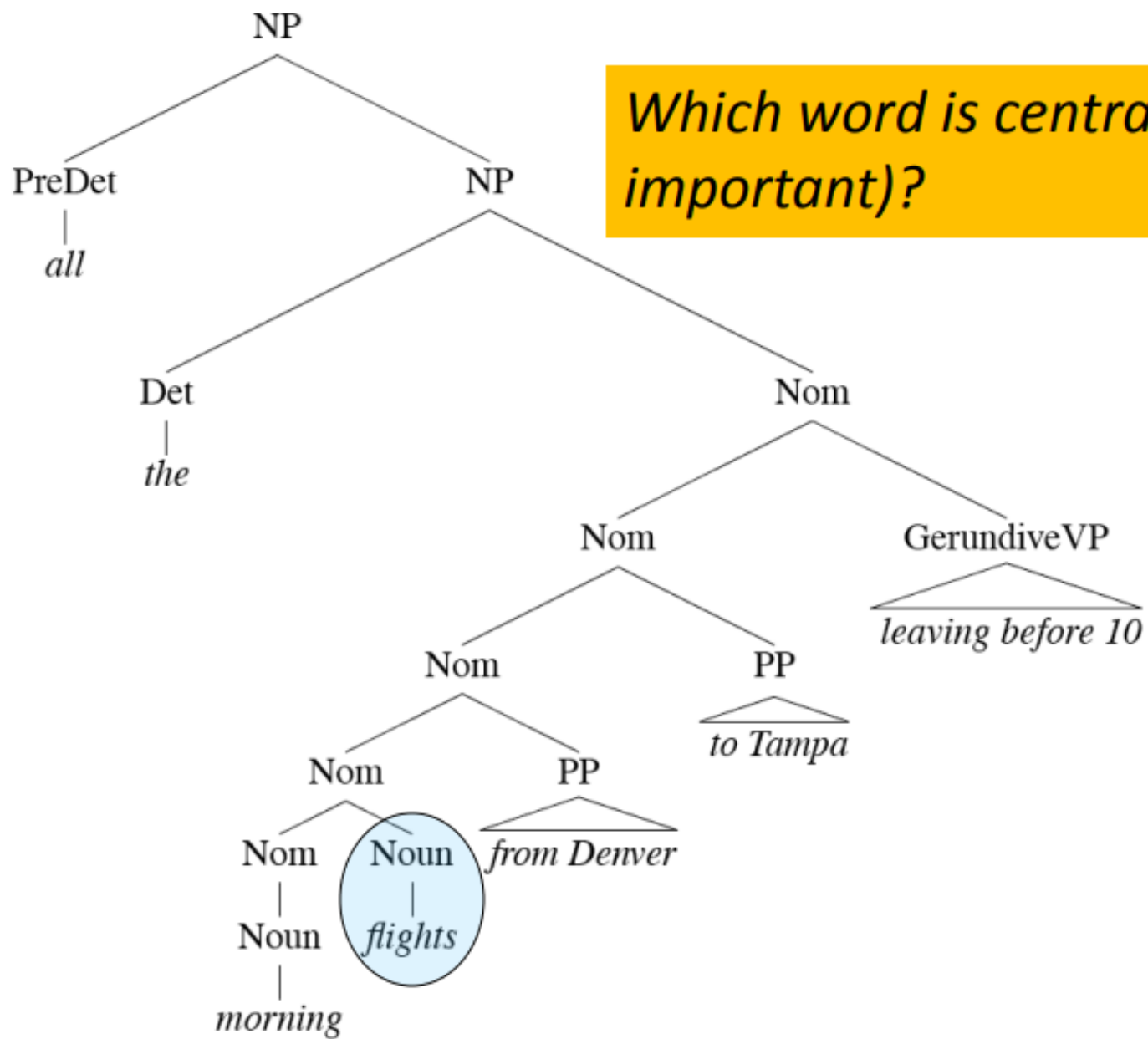
- Parsing is the process of taking a string and a grammar and returning a (or multiple) parse tree(s) for that string
- It is completely analogous to running a finite-state transducer with a tape
- It's just more powerful there are languages we can capture with CFGs that we can't capture with finite-state machines.

All the morning
flights from
Denver to Tampa
leaving before 10





Which word is central (most important)?



Which word is central (most important)?

NOUN PHRASE

- The noun phrase can be viewed as revolving around a head, the central noun in the noun phrase.
- The syntax of English allows for both
 - Prenominal (prehead) modifiers
 - Post-nominal (post-head) modifiers

Before the Head Noun

- Noun phrases can begin with a determiner, as follows:
a stop, **the** flights, **that** fare, **this** flight, **those** flights,
any flights, **some** flight.

CONTD...

- There are certain circumstances under which determiners are optional in English.
- For example, determiners may be omitted if the noun they modify is plural:

“Show me flights from San Francisco to Denver on weekdays”

- Word classes that appear in the NP before the determiner are called **predeterminers**.
- Many of these have to do with number or amount;
common predeterminer is all:

all the flights

all flights

CONTD...

- A number of different kinds of word classes can appear in the NP between the determiner and the head noun (the 'post-determiners').
 - **Cardinal numbers** – Eg. two friends, one stop
 - **Ordinal numbers** - include **first, second, third**, etc. but also words like **next, last, past, other, and another**
Eg. the first one, the next day, the second leg, the last flight, the other American flight, any other fares.
 - **Quantifiers** - **many, (a) few, several** occur only with plural count nouns: Eg. many fares.
 - The quantifiers **much and a little** occur only with noncount nouns.

NP STRUCTURE

- All the morning flights from Denver to Tampa leaving before 10 Clearly this NP is really about **flights**.
- That's the central critical noun in this NP. Such word is called as the **head**.
- We can dissect this kind of NP into the stuff that can come before the head, and the stuff that can come after it.

DETERMINERS

- Noun phrases can start with determiners...
- Determiners can be
 - Simple lexical items: the, this, a, an, etc.
 - A car
 - Or simple possessives
 - John's car
 - Or complex recursive versions of that
 - John's sister's husband's son's car

CONTD..

Adjectives occur after quantifiers but before nouns.

- a *first-class* fare
 - a *nonstop* flight
 - the *longest* layover
 - the *earliest* lunch flight
- Adjectives can also be grouped into a phrase called an adjective phrase AP. APs can have an adverb before the adjective

Eg. the least expensive fare

All the options for prenominal modifiers are combined with one rule as follows:

NP - (Det) (Card) (Ord) (Quant) (AP) Nominal

the use of parentheses () to mark optional constituents.

POSTMODIFIERS

A head noun can be followed by Post Modifiers.

Three kinds

- Prepositional phrases
 - Flights from Seattle
- Non-finite clauses
 - Flights arriving before noon
- Relative clauses
 - Flights that serve breakfast

POSTMODIFIERS

- any stopovers [*for Delta seven fifty one*]
- all flights [*from Cleveland*] [*to Newark*]
- arrival [*in San Jose*] [*before seven p.m*]
- a reservation [*on flight six oh six*] [*from Tampa*]
[*to Montreal*]

Here's a new NP rule to account for one to three PP postmodifiers:

Nominal : Nominal PP

POSTMODIFIERS

- The three most common kinds of non-finite postmodifiers are the gerundive (-ing), -ed, and infinitive forms.
- Gerundive postmodifiers are so-called because they consist of a verb phrase that begins with the gerundive (-ing) form of the verb.

CONTD...

- In the following examples, the verb phrases happen to all have only prepositional phrases after verb.
 - any of those (*leaving on Thursday*)
 - any flights (*arriving after eleven a.m*)
 - flights (*arriving within thirty minutes of each other*)

The use of a new nonterminal GerundVP:

Nominal : Nominal GerundVP

CONTD...

Rules for GerundVP constituents by duplicating all of our VP productions, substituting GerundV for V.

GerundVP - GerundV NP

GerundV PP

GerundV

GerundV NP PP

- GerundV can then be defined as:
- GerundV - being/preferring/arriving /leaving/ ...

CONTD...

A postnominal relative clause (more correctly a restrictive relative clause), is a clause that often begins with a relative pronoun (that and who are the most common).

The relative pronoun functions as the subject of the embedded verb in the following examples:

- a flight *that serves breakfast*
- flights *that leave in the morning*
- the United flight *that arrives in San Jose around ten p.m.*
- the one *that leaves at ten thirty five*

CONTD...

Rules like the following to deal with these

Nominal : Nominal RelClause

RelClause: (who / that) V P

CONTD...

Various postnominal modifiers can be combined, as the following examples:

- a flight *(from Phoenix to Detroit) (leaving Monday evening)*
- I need a flight *(to Seattle) (leaving from Baltimore) (making a stop in Minneapolis)*
- evening flights *(from Nashville to Houston) (that serve dinner)*
- a friend *(living in Denver) (that would like to visit me here in Washington DC)*

AGREEMENT

- Constraints that hold among various constituents.
- For example, in English, determiners and the head nouns in NPs have to agree in their number.
- Which of the following cannot be parsed by the rule
- NP → Det Nominal ?

(O) This flight

(X) This flights

(O) Those flights

(X) Those flight

CONTD...

- Which of the following cannot be parsed by the rule
- NP → Det Nominal ?
- This rule does not handle agreement! (The rule does not detect whether the agreement is correct or not.)

(O) This flight

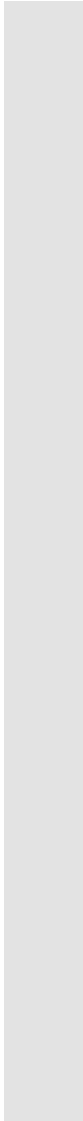
(X) This flights

(O) Those flights

(X) Those flight

PROBLEM

- Our earlier NP rules are clearly deficient since they don't capture the agreement constraint
- NP : Det Nominal
- Accepts, and assigns correct structures, to grammatical examples (this flight)
- But its also happy with incorrect examples (*these flight)
- Such a rule is said to **overgenerate**



THE VERB PHRASE AND SUBCATEGORIZATION

VERB PHRASE

- The verb phrase consists of the verb and a number of other constituents - **arguments**

$VP \rightarrow Verb$ disappear

$VP \rightarrow Verb NP$ prefer a morning flight

$VP \rightarrow Verb NP PP$ leave Boston in the morning

$VP \rightarrow Verb PP$ leaving on Thursday

SUB- CATEGORIZ- ATION

- But, even though there are many valid VP rules in English, not all verbs are allowed to participate in all those VP rules.
- We can subcategorize the verbs in a language according to the sets of VP rules that they participate in.
- This is a modern take on the traditional notion of transitive/intransitive.
- Modern grammars may have 100s or such classes

CONTD...

- Sneeze: John sneezed
- Find: Please find [a flight to NY]_{NP}
- Give: Give [me]_{NP}[a cheaper fare]_{NP}
- Help: Can you help [me]_{NP}[with a flight]_{PP}
- Prefer: I prefer [to leave earlier]_{TO-VP}
- Told: I was told [United has a flight]_S

CONTD...

- *John sneezed the book
- *I prefer United has a flight
- *Give with a flight

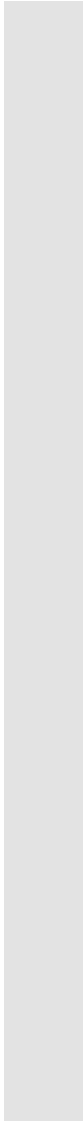
As with agreement phenomena, we need a way to formally express the constraints!

SUBCATEGORIZATION FRAMES AND EXAMPLE VERBS

Frame	Verb	Example
$\emptyset$	eat, sleep	I want to eat
NP	prefer, find, leave,	Find the flight from Pittsburgh to Boston
$NP\ NP$	show, give	Show me airlines with flights from Pittsburgh
$PP_{\text{from}}\ PP_{\text{to}}$	fly, travel	I would like to fly, from Boston to Philadelphia
$NP\ PP_{\text{with}}$	help, load,	Can you help [NP me] [NP with a flight]
VP_{to}	prefer, want, need	I would prefer [VP_{to} to go by United airlines]
VP_{brst}	can, would, might	I can [VP_{brst} go from Boston]
S	mean	Does this mean [S AA has a hub in Boston]?

CONTD...

- The various rules for VPs **overgenerate**.
- They permit the presence of strings containing verbs and arguments that don't go together
- For example : VP $\rightarrow$ V NP
- therefore *Sneezed the book* is a VP since “sneeze” is a verb and “the book” is a valid NP



GRAMMAR EQUIVALENCE & NORMAL FORM

GRAMMAR EQUIVALENCE

- A formal language is defined as a (possibly infinite) set of strings of words. Two kinds of grammar equivalence:
 - Weak equivalence
 - Strong equivalence
- Two grammars are strongly equivalent - generate the same set of strings and assign the same phrase structure to each sentence (allowing merely for renaming of the non-terminal symbols).
- Two grammars are weakly equivalent - generate the same set of strings but do not assign the same phrase structure to each sentence

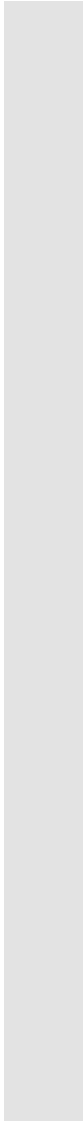
NORMAL FORM

- It is sometimes useful to have a normal form for grammars, in which each of the productions takes a particular form.
- For example a context-free grammar is in Chomsky normal form (CNF)
- Any grammar can be converted into a weakly-equivalent Chomsky normal form grammar.
- For example a rule of the form

$$A \rightarrow B C D$$

- can be converted into the following two CNF rules:

$$A \rightarrow B X$$
$$X \rightarrow C D$$



FINITE STATE & CONTEXT-FREE GRAMMARS

CONTD...

- Finite-state models of grammar are often inadequate - complex model of grammar
- The details of the syntax of noun phrases, we are prepared to discuss another problem with finite-state grammars.
- This problem is **recursion**.
- Recursion in a grammar occurs when an expansion of a non-terminal includes the non-terminal itself,
- Rules like **Nominal - Nominal PP**

CONTD...

Model the noun phrase up to the head with a regular expression (= FSA)

(Det) (Card) (Ord) (Quant) (AP) Nominal

- What about the postmodifiers? Let's just try adding the PP. We could then augment the regular expression as follows:

(Det) (Card) (Ord) (Quant) (AP) Nominal (PP)*

- So to complete this regular expression we just need to expand inline the definition of PP, as follows:

(Det) (Card) (Ord) (Quant) (AP) Nominal (P NP)*

CONTD...

- But wait; our definition of NP now presupposes an NP! We would need to expand the rule as follows:

(Det) (Card) (Ord) (Quant) (AP) Nominal (P (Det)
(Card) (Ord) (Quant) (AP) Nominal (P NP))*

The problem is that NP is a recursive rule.

CONTD...

- There is actually a sneaky way to ‘unwind’ this particular right-recursive rule in a finite-state automaton.
- In general, however, recursion cannot be handled in finite automata, and recursion is quite common in a complete model of the NP (for example for RelClause and GerundVP, which also have NP in their expansion):

(Det) (Card) (Ord) (Quant) (AP) Nominal
(RelClause / GerundVP / PP)*

CONTD...

- Chomsky (1959a) proved that a context-free language L can be generated by a finite automaton if and only
- if there is a context-free grammar that generates L that does not have any **center-embedded recursions** (recursions of the form $A \rightarrow \alpha A \beta$).
- The finite-state automata called the **recursive transition network or RTN** that adds the complete power of recursion to the FSA.
- The resulting machine is exactly isomorphic to the context-free grammar.